Avinash Agarwal

Civilization, AI, Healthspan, Space

Distributed Systems and Artificial Intelligence

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Summary

Seasoned Software Engineer with **8+ years of industry-leading experience at Meta and Microsoft**, passionate about tackling civilizational challenges, especially Healthspan and Space exploration! Proven track record of applying a research-driven approach to architecting highly available distributed systems processing billions of requests, leading cross-functional teams, and driving multi-million-dollar cost optimizations.

Experience

Meta

August 2025 — September 2026

Full-time Software Engineer (supervised by Gary G. Liu, Bill Hamilton, & Yun Zhang)

Menlo Park, California, USA

- **Applied AI & LLM Evaluations:** Drafted into the Applied AI organization to create model-evaluation tasks for Meta's next-gen LLM codenamed Avocado, under Meta's Superintelligence Labs.
- **AI for Healthspan:** Authored a visionary project proposal within the Applied AI org to process vast genetic datasets; advancing milestones like Google's AlphaFold, the proposal leverages modern AI to establish novel health benchmarks and accelerate cures for aging, rare-diseases, etc.
- **Engineering Productivity:** Root-caused an issue around pull-request insights where reviewer delays were incorrectly and entirely attributed to authors; filed a high-priority bug to ensure accuracy and fairness in developer analytics for company-wide engineering productivity.
- **Stability Lead**: Championed platform reliability operations for the unified authentication infrastructure, by actively triaging and resolving critical alerts, identifying and filing tasks for key SLI/SLO coverage gaps, systemic propagation delays around alert resolution, etc. (w/ Xinyi Zhang)
- **WhatsApp-based Authentication:** Authored a technical proposal to prioritize secure code-send channels while projecting multi-million dollars in cost-savings; fixed critical bugs around contact eligibility and internal server semantics; triaged further security improvements, video-call discrepancies, etc.
- **Meta.ai Platform Reliability:** Stabilized login flows for the Meta.ai application by systematically resolving complex, intermittently failing cross-platform test suites, significantly improving CI/CD reliability.
- **On-Call Agent:** Enhanced an alert summarizer agent by resolving prompt completeness errors; collaborated directly with the company-wide on-call platform owners to include this agent in their workflow, significantly improving engineering productivity (w/ Nanway Chen)
- **On-Call Operations & Incident Management:** Completed specialized onboarding; monitored dashboards, triaged alerts, completed reports, proactively root-caused a privacy anomaly, and collaborated with relevant owners to improve time-to-mitigate and prevent future outages.
- **Secured the runner-up position in an internal Hackathon:** Demonstrated high engineering velocity by rapidly fixing a high volume of issues while triaging improvements for the hackathon platform.
- **Dogfooding & Virtualization:** Supported on-device dogfooding for unified login flows (w/ Nanway Chen) and ramped-up on internal virtualization frameworks for testing most scenarios virtually (w/ Feng Wan)
- **Global Identity Architecture & Execution:** Led the end-to-end design and implementation of 8+ distinct account-recovery eligibility methods; instrumented relevant telemetry validating the safe transitioning of billions of requests across the Meta ecosystem (w/ Zach Yang, Feng Wan, Yuxi Tao, etc.)
- **Observability & Quota Management:** Audited telemetry sources to resolve persistent capacity issues; optimized sampling rates and created distinct logging partitions for partner teams (w/ Mayank Chacko)
- **Privacy-Safe Infrastructure:** Modernized legacy components to use privacy-safe wrappers while onboarding to the unified authentication infrastructure team.

- **Infrastructure Bootcamp:** Dedicated almost a month gaining hands-on training to successfully onboard into my officially hired role on **Infrastructure Engineering** at Meta.

Microsoft Azure Core

March 2022 — July 2025

Full-time Software Engineer (supervised by Gaurav Kapila, Chirag Gupta & Jenny Hunter)

Redmond, Washington, USA

- **Azure Resource Graph**
  - **Engineered data ingestion components** hosted on the Azure Synapse platform to support dynamic time-window loading for security datasets (w/ Sanjai Ganesh Jeenagiri, Minerva Chen, & Umair A.)
  - **SQL Query Optimization:** Optimized complex queries executed on Azure Synapse to efficiently load and merge concurrent incoming data streams. (w/ Sanjai Ganesh Jeenagiri, Sivaraman Venkataraman, Abhinav Gupta, etc.)
  - **Championed increasing partition-key cardinality**, leveraging data-driven demonstrations to root-cause and resolve systemic skewness across the entire stack (w/ Jing Fang)
  - **High-Performance (de)serialization:** Independently root-caused bottlenecks via profiler analysis and championed the stack-wide adoption of System.Text.Json, leveraging its faster, lower-allocation APIs to optimize payload processing at scale (w/ John Call)
  - **Buffered Encryption:** Implemented buffered streaming over an existing encryption algorithm to drastically reduce active memory allocations for massive strings, while ensuring identical hash outputs and downstream compatibility (w/ Alan Yang, John Call, & Olha Tkachenko)
  - **Telemetry & Partition Observability:** Instrumented granular partition-specific metadata across Event Hub ingestion streams, substantially improving time-to-detect and root-cause analysis for relevant partition-level anomalies (w/ Olha Tkachenko)
  - **Profiled and optimized services** eliminating contention bottlenecks and achieving nearly 10% improvement in compute, memory and network efficiency.
  - **On-Call Operations:** Triaged, mitigated, and root-caused live production incidents during scheduled rotations and actively improved efficiency through multiple enhancements to internal monitors, troubleshooting-guides, etc. (w/ Pooja Narayanan, Pinwen Su, Monica Mihailescu, etc.)
- **Azure Resource Change Analysis**
  - **Streaming Pipeline (1B+ daily notifications) Migration:** Contributed to an approved transitioning design-doc and executed foundational steps to migrate the high-throughput streaming data pipeline for Change Analysis to its dedicated scale unit (w/ Abhinav Gupta, Harkirat Singh & Daniel Hu)
  - **Change Safety Integration:** Guided partner teams on integrating with Change Analysis resources, authoring and deploying API/Swagger specifications while reviewing inbound pull requests to ensure cross-stack architectural safety (w/ Ayaz Ali & Bharath Kalidindi)
  - **Front-End General Availability:** Worked on improving performance Lighthouse scores, deployments and test automation, completing accessibility fixes, etc. (w/ Hardhik Mallipedi, Zhonghua Bao & Julia Wang)
  - **Proxy Resource Change Tracking:** Contributed to the design and deployment of Change Analysis backend services for Azure proxy resources, expanding auditing coverage and closing visibility gaps across nested topologies (w/ Daniel Hu)
  - **Change Actor General Availability**: Designed and presented a comprehensive design doc on a new compliance and auditing functionality enabling customers to audit WHO initiated a change and WHICH client interface (Portal, CLI, SDK, etc.) was used to make these changes (w/ Aydan Yumerefendi). Worked on end-to-end implementation, cross-team collaboration and project completion (w/ Amit Dua, Virat Satyam, & Daniel Hu)
  - **Architectural Evolution:** Championed selective caching of internal columns informed by customer query telemetry (w/ Hardhik Mallipedi) to significantly reduce active memory footprint; worked on end-to-end removal of nullable fields, ensuring data integrity and API/Swagger contract migration to dramatically reduce footprint under strict resource constraints (w/ Jaeyoung Yoon)

- **Partition Skewness Reduction:** Optimized data-ingestion pipelines, achieving over 30% improvement in compute, memory and network efficiency (w/ Jing Fang)
- **Fast-Path Rollout:** Contributed to the design and staged rollout of fast-path execution pipelines for Change Analysis, unlocking massive throughput improvements; authored extensive test suites to ensure correctness, before full rollout (w/ Jaeyoung Yoon)
- **Backend General Availability:** Deployed relevant services, dependencies, etc. across US and China Government clouds, partnered with authorized personnel for sovereign air-gapped clouds, timely achieving critical compliance milestones directly supporting Azure's enterprise market positioning.
- **On-Call Operations:** Triaged, mitigated, and root-caused live production incidents during scheduled rotations and actively improved efficiency through multiple enhancements to internal monitors, troubleshooting-guides, etc. (w/ Abhinav Gupta, Daniel Hu, Jaeyoung Yoon, etc.)

Microsoft Azure Global

June 2018 — February 2022
Full-time SWE (supervised by Subrata Roychoudhuri, Arun Ramamurthi, & Saumya Chandra) Hyderabad, Telangana, India

- **Microsoft Community Training** — Worked on core development of an Azure-powered learning platform incubated out of Microsoft Research (MSR) India to train the next billion users— anchored on the thesis around education being a foundational civilization mechanism to navigate The Great Filter.
  - **Hyperscale Platform Modernization**
    - **Analytics Architecture Revamp:** Authored a technical proposal outlining the addition of a pull-based analytics model and asynchronous workflows to resolve systemic data integrity bugs; drove foundational improvements by implementing the pull model and executing extensive query optimizations, significantly improving platform performance for learner workflows.
    - **Performance Profiling & Load Testing:** Performed load testing and performance profiling using Apache JMeter, ensuring platform stability for massive-scale deployments totaling over 10M+ learners and 1B+ course-completions (verified by Garima Kamthan)
    - **Azure Load Testing & Terraform:** Identified and merged a fix significantly improving Azure-Load-Testing's support in the multi-cloud cohort supported by Terraform.
    - **High-Scale Data Migration:** Architected APIs to successfully execute the migration of 160M+ course-completion records from another major cloud provider to Azure (w/ Nidhi Kothari)
    - **Authentication Load Testing:** Designed and implemented scripts to locally sign JWT tokens for end-to-end load-testing of authentication flows (w/ Kinshuk Singh)
    - **Throughput Optimization:** Achieved RPS improvements of up-to **20x** across core APIs and **10x** across several others; analyzed execution plans and fixed SQL queries, eliminated redundant network calls and implemented strategic caching layers (w/ Kinshuk Singh)
    - **Architectural Leadership:** Championed data-store scale-out strategies including SQL Server read replicas (w/ Kinshuk Singh) and distributed caching with Redis to minimize long-term technical debt while balancing rapid iteration requirements.
  - **UNICEF Learning Passport** (supervised by Sandhya Guntreddy)
    - **Offline Innovation (Hackathon Lead):** Spearheaded a hackathon project to host the entire cloud service locally on Windows machines. The platform was recognized as one of **TIME's Best Inventions of 2021** for providing continuous education to children in conflict zones and refugee settings (w/ Alok Agrawal)
    - **Global Societal Impact:** This offline-first prototype was the key technical catalyst that prompted UNICEF to launch the initiative using Microsoft Community Training for no-connectivity regions, contributing to a high-scale initiative that reached **7.4 million learners across 43 countries** by 2024.
    - **Strategic Architectural Leadership:** Championed and worked on containerization and migration to cross-platform .NET Core; this move was critical for the platform's ability to run on evolving hardware in emerging markets (w/ Vishal Goyal & Sanmoy Ray)

- **Operational Support & Edge Strategy:** Worked on Azure IoT Hub integration to support remote management and observability for offline learning devices; assisted in testing data-sync to ensure reliable curriculum delivery in offline settings (w/ Vishal Goyal, Kinshuk Singh, & Sanmoy Ray)

- **Azure Marketplace Public Preview Launch**
  - **Platform Architecture & Onboarding:** Ramped-up on Azure Managed Applications, and presented the deployment workflow to an Azure Global leadership audience (w/ Vishal Goyal)
  - **Automated Marketplace Deployment:** Engineered robust end-to-end deployment automation workflows to provision and configure all foundational cloud resources, services, and dependencies directly within customer Azure tenants (w/ Vishal Goyal, Brian Moore & Sukhjeevan Reddy)
  - **Azure Portal UX Templates:** Designed and implemented the complete suite of Azure Resource Manager (ARM) UI definitions and portal templates (‘createUiDefinition.json‘), streamlining initial parameter configuration and onboarding UX for platform customers (w/ Vishal Goyal, Gary K. & Deepak Kathayat)
  - **Architectural Leadership:** Evaluated and formally championed Azure Lighthouse to resolve cross-tenant management constraints, demonstrating how Lighthouse’s would have accelerated time-to-market and eliminated multi-tenant architectural debt across Azure Global.

- **Swachh Bharat Mission**
  - **Mobile Engineering Leadership:** Directed an engineering team through the end-to-end implementation of key mobile features, culminating in the successful launch of the official Swachh Bharat Mission eLearning mobile app (w/ Abhinav Sobti)
  - **Rapid Triage & Migration:** Spearheaded the rapid triage of launch-critical issues w/ Aditya Agarwal, product and program management teams etc. to ensure a smooth migration of the training platform.
  - **Xamarin UI Architecture:** Studied, architected and timely implemented a high-impact search-enabled dropdown in Xamarin to resolve a critical UI bug; this allowed users to efficiently navigate and select from the hundreds of cities where the platform was deployed.
  - **National Social Impact:** Contributed to a high-scale social impact project that successfully trained and empowered over **110,000+ municipal functionaries** across **4,000+ cities** in India.

- **Platform Search Infrastructure** (w/ Alok Agarwal)
  - **Data Ingestion & Indexing:** Engineered workflows using Azure Cognitive Search to ingest data residing across SQL and Blob Storage into a custom search index.
  - **Search Relevance & Retrieval:** Featured Lucene-based retrieval and implemented search relevance enhancements leveraging pretrained semantic analyzers (e.g., fuzzy matching, edit distance, synonyms, phonetics).
  - **Unstructured Content Extraction:** Used Azure Cognitive Services to extract and process content from unstructured file formats, including documents and videos.

- **Azure Front Door Integration & Performance Optimizations**
  - **Mission-Critical BCDR:** Designed and implemented a Load Balancing and Disaster Recovery architecture using Azure Front Door (already benchmarked by LinkedIn)
  - **Edge Security:** Significantly hardened the platform’s security posture by configuring web application firewalls, DDoS protection etc. at the edge.
  - **Custom Domain Configuration:** Modernized legacy custom domain configuration procedures, streamlining customer onboarding and SSL/TLS certificate management.
  - **Performance Optimization:** Benchmarked and implemented API response compression (Brotli, Gzip, Deflate, etc.) to significantly improve page-load latencies.
  - **Caching Strategy:** Implemented CDN-based caching of static assets and tuned CDN and browser caching configurations (Cache-Control) for relevant APIs.
  - **Large-Scale Migration:** Orchestrated the seamless migration of 100s of customers to the new Azure Front Door endpoints; involved careful cross-functional planning and execution.

- **Platform's Identity & Authentication Infrastructure**
  - **Multi-Region Resilience & BCDR:** Architected and implemented multi-region IdentityServer deployments behind Azure Front Door and configured active-active global load balancing and zero-downtime disaster recovery.
  - **Data Tier Decoupling & Migration:** Migrated tenant-specific deployment metadata including different configurations and token-signing certificates from SQL Server to globally replicated Azure Cosmos DB to enable low-latency global token verification (w/ Sanmoy Ray)
  - **.NET Core Modernization:** Contributed to migrating the platform's core identity infrastructure including the IdentityServer components to .NET Core, while enabling container-ready execution across cross-platform environments (w/ Vishal Goyal & Saloni Chandak)
  - **Auth Provider & Cost Optimization:** Spearheaded the migration of SMS based code-send provider to dedicated third-party infrastructure; collaborated directly with Telesign and engineered the end-to-end integration, drastically reducing worldwide code-send costs.
- **Platform Localization**
  - **Leadership:** Engineered a scalable internal localization framework and integrated with company-wide manual translation pipelines, eventually leading accessibility efforts across the entire platform, while keeping most of it extensible across Azure Global.
  - **RTL & Complex Scripts:** Delivered comprehensive right-to-left (RTL) format support and resolved complex ligature rendering, ensuring accurate typography across the platform.
  - **Custom PDF Engine:** Architected a templatized HTML-to-PDF certification feature with strict security and localization constraints; partnered with IronPDF to significantly accelerate time-to-market.
  - **Live Demonstration:** Presented RTL capabilities to a large audience, gracefully utilizing live product errors to demonstrate comprehensive, end-to-end localization coverage.
- **Platform Observability:** Worked on implementing a highly scalable architecture using Event Hubs and Azure Data Explorer (w/ Vishal Goyal) to aggregate telemetry across managed deployments. Utilized cross-tenant Application Insights queries and Managed Identities with AMA access to enable diagnostic and analytic dashboards (w/ Deeksha Shravani)
- **Cross-Platform Test & Release Automation:** Engineered release pipelines to concurrently compile customer-branded Android packages in parallel; integrated Microsoft's App Center to orchestrate test runs across heterogeneous device matrices, mitigating flakiness, accelerating execution, and scaling device coverage with an expanded test plan (w/ Venkat Kancheti)
- **API Test Automation:** Worked on comprehensive functional test suites using Postman and Newman, specifically scripting multi-step authentication handshakes, dynamic session token propagation etc.
- **Telemetry & Growth Analytics:** Integrated Application Insights with advanced funneling to track end-to-end learner journeys from landing to completion; configured a Dynamics 365 integration with Azure Marketplace to capture and generate sales leads based on platform telemetry (w/ Deepak Kathayat)
- **Platform Health probes:** Architected and implemented health-check endpoints instrumenting subsystem diagnostics for core dependencies including database connectivity, blob storage availability, certificate expiration telemetry, etc. significantly improving time to detect and mitigate outages.
- **Platform Versioning & Upgrades:** Introduced an aggregated platform versioning framework to govern system-wide release cycles; automated deployment-time execution of existing database migration scripts and relevant cross-service dependency updates.
- **Media Processing Reliability:** Resolved critical state-tracking synchronization failures across video encoding workflows by diagnosing, implementing, and rigorously testing background cron-jobs orchestrating with Azure Media Services (w/ Vishal Goyal & Anchal Goyal).
- **DRM & Content Protection:** Authored a technical proposal to modernize content protection, championing the use of industry-standard DRM protocols (Widevine, PlayReady, and FairPlay) beyond just AES-128 encryption.

- **High-Throughput Bulk Ingestion:** Architected and built an end-to-end bulk upload system enabling parallel ingestion of multi-format media; significantly reduced administrative onboarding time and super-charged the manual testing experience for everyone.
- **Dependency Graph Optimization:** Identified and eliminated cyclic dependencies across interdependent .NET project references; decoupled shared assemblies and streamlined dependency graphs, reducing overall package-size footprint **by over 70%**, dramatically improving downstream workflows like build and deployment runtimes and overall developer velocity. (w/ Mayukh Chakraborty)
- **Threat Modeling:** Conducted deep-dive attack surface analysis using Microsoft internal security frameworks (w/ Vishal Goyal); ensuring the security and integrity of user data and platform content.
- **Privacy & Compliance:** Enabled automated compliance tooling to identify secure and compliant dependencies; ensuring rigorous adherence to global data protection and privacy standards.
- **Deployment & Test Automation:** Engineered daily automated deployment pipelines across all internal development and test environments; orchestrated the execution of comprehensive test suites post-deployment, significantly accelerating developer velocity and overall testing productivity.
- **Offline Content & Sync Reliability:** Diagnosed and resolved critical offline content caching and learner progress synchronization issues within the Xamarin mobile application, ensuring reliable curriculum access for users in regions with zero or intermittent connectivity (w/ Kulwinder Kaur)
- **Error Diagnostics & Triage Automation:** Conducted a comprehensive deep-dive into service and client layer error handling, aggregating telemetry into Application Insights and establishing automated daily alerts prioritizing high-impact regressions.
- **State Persistence & User Analytics:** Engineered full-stack features for real-time tracking of user progress, including most-recently-viewed-lesson and last-watched-timestamp; involved REST API and database schema updates to record relevant events, and frontend enhancements across web and mobile to resume content appropriately.
- **Platform Notifications Infrastructure**: Implemented client-side Xamarin APIs (w/ Rohan Badlani) before leading the notification subsystem for the entire platform (w/ Aman Gour); supported the paginated, pull-based service and evaluated Azure Notification Hubs for real-time push notifications.
- **Mentorship & Technical Review:** Reviewed project specifications and technical design documents for mentee interns and peers.
- **On-Call Operations:** Triaged, mitigated, and root-caused live production incidents during scheduled rotations and actively improved efficiency through multiple enhancements to internal monitors, troubleshooting-guides, etc. (w/ Aditya Agarwal, Mayukh Chakraborty, Vidhu Mangal, Anchal Goyal, Saloni Chandak, Deepak Kathayat, Abhinav Sobti, etc.)

Microsoft Garage Hackathons

June 2018 — July 2025

Engineer

Hyderabad, Telangana, India

- **Azure Oracle:** Conceptualized a context-aware recommendation engine leveraging Change Analysis datasets to suggest next-best-actions for Azure cloud workflows.
- **LLM-based API Governance:** Prototyped a prompt-engineering proposal (w/ Robin Chapas) to automate the categorization of Azure Swagger API properties (security, performance, costs, etc.)
- **Kool AI(d) for Gamers:** Fine-tuned pre-trained computer vision models (w/ Shivam Tyagi) to extract real-time HUD metrics and game states from live Dota-2 streams. The collaborative team effort spanning around 100 individuals secured 2nd place across the entire company.
- **Touchless Kiosk Ordering:** Participated in a project prototyping a computer vision gesture-recognition interface for contactless ordering on self-service kiosks (w/ Vivek Shankar, Kaustabh Kakoty, etc.)
- **Koev Halev:** Actively contributed at the IDC campus, providing domain expertise in cross-functional design discussions for a holistic project focused on mental and emotional well-being.
- **Sustainable Cloud Provisioning:** Participated in a project prototyping a solution to explicitly highlight carbon-efficient Azure SKUs, thus incentivizing carbon footprint reduction.

- **Web3 Compute Utilization:** Participated in a project proposal to utilize idle Azure compute to support blockchain networks, providing resources to operate cryptographic mining and validation nodes.
- **Immutable Voting Ledger:** Participated in a project championing the use of cryptographic blockchain frameworks to ensure immutability and overall integrity in democratic elections.
- **IoT Home Automation:** Collaborated within a team (w/ Abhishek Raj, Shivam Tyagi, etc.) to build a Raspberry-Pi powered IoT automation layer to wirelessly control legacy electric switches retrofitted with motorized 3D-printed actuators.

Microsoft Code.Fun.Do
2018 — 2019
Engineering Lead and Pizza Logistics Moderator (supervised by Vijay Vavilala & Sandeep Chadda)
India

- **Event Moderation & Algorithmic Track Proposal:** Delivered talks on technical-guidelines, project-submissions, etc; managed ad-hoc troubleshooting and Pizza logistics at NIT Surathkal and BITS Pilani (Hyderabad); championed supplementing competitive project-based events with structured problem-solving contests (Google Kickstart, Google Code Jam, Meta Hacker Cup, ACM-ICPC, etc.) positioning rigorous algorithmic proficiency as a **foundational pillar for engineering talent.**

Goldman Sachs
September 2016 — October 2016
Team Lead, Software Engineering Internship
Bengaluru, Karnataka, India

- **National Runner-Up (GS Quantify 2016):** Led a 3-member team in this flagship interdisciplinary event, serving as the primary contributor for the advanced algorithmic problem-solving track.
- **Data Science**: Contributed to the data-science track for bond-liquidity-prediction (w/ Samiksha Gupta)
- **Quant**: Co-directed the quantitative solution track to finalize the project submission (w/ Shivam Tyagi)
- **Executive Presentation:** Earned a fully sponsored invitation to their Bengaluru headquarters (hosted at the prestigious Hilton Embassy) where the team delivered the concluding presentation.

Microsoft Foundry
May 2016 — July 2016
Software Engineering Internship (supervised by Pankaj Garg & Saumya Chandra)
Hyderabad, Telangana, India

- **Conversational AI:** Engineered chat-bots using the **Microsoft Bot Framework**, architecting custom templatized conversational flows for form-filling, within a cross-functional team of 8 members.
- **Computer Vision:** Processed images using OpenCV to better highlight information, significantly improving downstream OCR engine performance.
- **Natural Language Processing:** Tested and evaluated experiences built with Microsoft LUIS.
- **Live Technical Demonstration:** Delivered the final product presentation to a large Microsoft Foundry audience, showcasing the conversational flows for form-filling.

Xerox Research Centre
Dec 2015 — Jan 2016
Research Internship (supervised by Vaibhav Rajan)
Bengaluru, Karnataka, India

- **Xerox Research Innovation Challenge:** Engineered neural network models, augmenting the provided patient vitals with synthetically generated data for online mortality prediction of ICU patients.
- Completed a Winter School on predictive data-science (w/ Dushyant Kumar & Sartaj Singh)
- Preprocessed massive unstructured lab-result datasets and built ensemble models combining them with patient vitals to predict both mortality and broader health complications.
- Secured **third place** (online-round) and **seventh place** (onsite-round) in the overall challenge; successfully competing against specialized research teams.

ACCOMPLISHMENTS

- **Google Kickstart Round D 2017:** Ranked **83rd** internationally.
- **Codeforces Round #410 (Div. 2):** Ranked **15th** internationally (2nd in India)
- **ACM-ICPC 2016** (w/ Umang Goel & Himadri Mishra): Qualified for and represented IIT (BHU) Varanasi at the Asia-Amritapuri regionals.
- **IndiaHacks 2016** (w/ Sartaj Singh): Secured a **14th rank** against 3,000+ other teams in their data-science track; invited to the on-site finals held at **Vivanta by Taj** in Bengaluru, KA.
- **ACM-ICPC 2015** (w/ Aditya Kapoor & Shivam Tyagi): Qualified for and represented IIT (BHU) Varanasi at the Asia-Amritapuri regionals.
- **CodeChef Long Challenge:** Achieved an overall India Rank of **31** in the Long Challenge track.
- **CodeChef May Challenge 2015:** Secured a **national rank of 19** in this Long Challenge.
- **IIT Bombay TechFest-15:** Ranked 25th in the International Coding Challenge Final Round.
- **HackerEarth October Sky:** Ranked **16th** among 1700 international candidates.
- **JCECE 2013 (Medicine): State Rank 1** in the Jharkhand Combined Entrance Competitive Examination!
- **JCECE 2013 (Engineering): State Rank 2** in the Jharkhand Combined Entrance Competitive Examination!
- **NEET 2013: All India Rank 6355** among 650k+ candidates in the National Eligibility cum Entrance Test.

PROJECTS

- **Steganography:** Implemented a system to securely embed and exchange hidden messages within digital images, enabling applications such as authenticated watermarking and media provenance.
- **Multi-Agent Environment Simulation:** Built a simulation of independent RL-Agents coordinating with each other to efficiently collect items distributed in a two-dimensional grid.
- **AIMA-CSharp Open Source Contribution:** Contributed to the AIMA-CSharp repository by implementing pseudo-code from Peter Norvig's **Artificial Intelligence: A Modern Approach** in C#.
- **The Chess Game:** Architected and developed a full-stack chess platform in Java with live multiplayer matchmaking and an interactive chatroom, subsequently extending it with an AI-driven single-player mode utilizing Minimax and Alpha-Beta pruning.
- **Rectifier:** Conceptualized circuit designs for full-wave AC-to-DC conversion and capacitive output filtering.
- **2D Physics Game Engine:** Built a 2D rigid-body simulation in Java featuring multi-object collision detection, kinematic interactions, and configurable gravity environments including near-Earth uniform acceleration and deep-space Newtonian $1/r^2$ dynamics.
- **Tic-Tac-Toe AI:** Built a multiplayer Tic-Tac-Toe game, subsequently extending it with rule-based single-player AI utilizing an if-else decision tree across two difficulty tiers, engineered to be unbeatable on hard.
- **Calendar Day Calculator:** Built a Java application to calculate the exact day of the week handling dates across millennia including logic to handle chronological edge cases.
- **CPU Overclocking**: Successfully overclocked an Intel Pentium E5200 processor from a base clock of 2.5GHz to 2.8GHz; involved tuning FSB frequencies, CPU multipliers, and voltage settings; benchmarked thermal output and system stability under peak loads.
- **Barometer:** Built a DIY pressure barometer using a sealed jar, a flexible rubber diaphragm, and a graduated scale to track ambient atmospheric pressure variations.

- **Water Level Controller:** Built a DIY system utilizing the electrical conductivity of tap water to detect when a tank is full, thus automating overflow protection.
- **Periscope:** Built a DIY prism-based periscope using right-angle glass prisms and a cardboard box housing to demonstrate total internal reflection and observe objects over obstacles.

Skills

- **Leadership** — Accountability, Collaboration, Prioritization, Alignment, Reviews, etc.
- **Software Engineering** — Data Structures and Algorithms, Distributed Systems (Billions of Daily Notifications), Event-Driven Architecture & Data Pipelines, Authentication and Identity Systems, Blockchain, API Design and Smart Contracts, IaaC, etc.
- **Networking** — TCP/UDP, Load Balancing, Reverse Proxies, CDN, HTTP/HTTPS, DNS, RPC, etc.
- **Data & Storage** — Redis, NoSQL, SQL, Blockchain, etc.
- **Cloud & Infrastructure** — Microsoft Azure (Expert), AWS, Kubernetes, CI/CD, Automation, etc.
- **Site Reliability** — Business Continuity and Disaster Recovery, On-call and Incident Response etc.
- **Artificial Intelligence** — Reinforcement Learning (Agentic Workflows, Exploration vs Exploitation tradeoffs, Play, etc.), Deep Learning (Neural Networks, CNNs, RNNs, Transfer Learning, etc.), Machine Learning (Mathematical Modeling and Search Heuristics), Data Science (Statistics, Classification, Regression, PCA etc.), LLM Fundamentals and Applications, etc.
- **Search & Retrieval** — Inverted Indexes, Semantic Analyzers, Apache Lucene, Elasticsearch, etc.
- **Frameworks & Libraries** — JVM, .NET, TensorFlow, Keras, Pandas, Scikit-learn, Caffe, XGBoost, OpenCV, Unity, React, ARM, IronPDF, NumPy, Sage, Selenium, etc.
- **Programming Languages** — Java, C#, C++, Python, TypeScript, PowerShell, Hack, etc.
- **Hardware** — Electronics, Gadgets, Device Assembly & Tuning, Firmware & Semiconductor Fundamentals.
- **Business** — Product Growth Analytics, Strategic Roadmapping, Enterprise Cost Optimizations etc.
- **Visual Arts** — UI/UX Typography & Localization, Digital Media Processing, Illustrations, etc.
- **Technical Writing** — Documentation, System Design, Incident Post-Mortems, Academic Research etc.
- **People** — Model, Coach, Care, Ally!
- **Security** — Access Control, Rate Limiting, Encryption, Threat-Modeling, CTF-events, etc.
- **Privacy** — PII, OII / CUI, Anonymization & Pseudonymization, Cryptographic Hashing, CCPA, GDPR, etc.
- **Natural Languages** — English, Hindi, Marwari (Intermediate), Sanskrit (Beginner), etc.

Education

coursera Coursera.org — April 2018 — Present
Specializations and Certifications — Online

- **DeepLearning.AI:** Generative AI For Software Development
- **The State University of New York:** Blockchain Basics and Smart Contracts
- **University of Alberta & Amii:** Reinforcement Learning Specialization
- **DeepLearning.AI:** Deep Learning Specialization

Indian Institute of Technology (Banaras Hindu University)

July 2013 — May 2018

Integrated Master's degree (5 years): Mathematics & Computing

Varanasi, Uttar Pradesh, India

- **Academic Performance:** Graduated with a Cumulative Grade Point Average (CGPA) of 8.7/10.0
- **Thesis:** Deep Learning based Classification of Handwritten Mathematical Symbols. (supervised by Tanmoy Som)
  - **Data Augmentation:** Diagnosed class imbalances in the original HASYv2 dataset of 369 symbols, and implemented a rotation augmentation pipeline for underrepresented symbols.
  - **CNNs & Transfer Learning:** Leveraged local NVIDIA GPUs to fine-tune modern convolutional neural network architectures (DenseNet, WideResNet, MobileNet, and NASNet) on the final dataset using transfer learning methodologies.
  - **Empirical Results:** Demonstrated that resolving data quality and class distribution imbalances reduced model variance and improved convergence, validating the critical need for both manually curated and synthetic datasets to achieve state-of-the-art accuracies.
  - **Thesis Defense:** Successfully defended the thesis through a detailed technical presentation covering CNN architectures, transfer-learning methodologies, and the final classification outcomes.
- **Campus Ambassador – Microsoft CODESS:** Partnered with Microsoft to champion inclusion in tech by leading the internal distribution of the CODESS launchpad program for female engineering students.
- **Teaching Assistant** for the Mathematical Methods course MA-203, offered by the Department of Mathematical Sciences (supervised by Manushi Gupta)
- **Problem Setter & Moderator, Technex–16:** Mathletics and Prayaas
- **CodeDef, Udyam–2016:** Ranked 1st in this coding contest organized by the Electronics branch.
- **Analiticity, Technex–16:** Ranked 2nd in this campus-wide data-science contest.
- **COPS-OPC–2015:** Ranked 1st in this algorithmic problem solving focused coding contest.
- **Microsoft Code.Fun.Do:** Worked on developing a 2D top-down space shooter game using the Unity game engine (w/ Sumit Sahrawat, Arun Meena, & Shivam Tyagi)
- **Matheletics, Technex–15:** Ranked 3rd in this Project Euler style Mathematics and Computing event.
- **Predicting Tourism Trends – DIH Project** (supervised by Santwana Mukhopadhyay) Surveyed domain literature and built statistical and machine learning models to forecast regional tourism trends using visitor records from regional tourism board portals and search volume histories from Google Insights (w/ Abhishek Bind & Arun Meena)
- **Rediscovering Darkness:** Stood 2nd in an Astronomy Quiz hosted by the Astronomy Club!
- **Treasure Hunt, KashiYatra–14:** Runner's up across all of Varanasi (w/ Abhishek Bind & Arun Meena)
- **Mod-Ex, Technex–14**: Ranked 2nd for building an open-source chess game (w/ Shivam Tyagi)
- **Robo-Soccer, Technex–13:** Runners-up (w/ Sumit Sahrawat, Rishabh Daal, etc.) out of 100+ teams at the Robotics-Club's event; conceptualized electronic circuit designs for a D-Pad controller to operate an All-Wheel-Drivetrain.
- **Event Organizer & Moderator**: Competitive tournaments for AoE-2, CoD-4, NFS, Civ-5, and DotA-2!

FIITJEE Forum for IIT-JEE

April 2011 — June 2013

Joint Entrance Examination

Dhanbad, Jharkhand, India

- **All India Rank 494** in Joint Entrance Examination — 2013, among 1.3 million candidates!
- **All India Rank 1672** in IIT-JEE (ADVANCED) — 2013, among 150k candidates!
- **Stood 1st** in the batch of 2013, among approximately 400 other candidates!

Delhi Public School

June 2011 — April 2013

Central Board of Secondary Education

Dhanbad, Jharkhand, India

- **Scored 96.5%** in the All India Senior School Certificate Examination (AISSCE) — 2013!
- **Awarded DST INSPIRE Science Scholarship for ranking in the top 1%** of AISSCE — 2013!
- Recognized in the **top 0.1% of AISSCE — 2013 for excellence in Physical Education!**
- Drafted into the **Aryabhatt** advanced mathematics program; competed in the school-level IMO round.
- **Practicals:** Biology, Chemistry, Physics.

De Nobili School, Sijua

March 2000 — March 2011

Indian Certificate of Secondary Education

Dhanbad, Jharkhand, India

- **Received 90%+ in FTRE Scholarship** in a national-level talent exam among 100k+ candidates!
- **Scored 87.9%** in the Indian Certificate of Secondary Education — 2011 examination.
- **Theory:** Mathematics, Language & Literature (English & Hindi), Physics, Chemistry, Biology, Computers, Environmental Sciences, History, Civics, Geography, etc.
- **Practicals:** Reading, Speaking, Writing, Physical-Training, Art & Craft, Computers, Chemistry, Biology, Physics, Topology, etc.

PURSUITS

- **Metaverse** — Brain Computer Interfaces, Spatial Computing, Augmented and Virtual Reality, etc.
- **Science** — Medicine, Anatomy, Physics, Chemistry, etc.
- **Culinary** — Foodie (Advanced), Cooking (Intermediate), etc.
- **Fitness** — Yoga (Beginner), Gym (Moderate), etc.
- **Sports** — RTS, FPS, CP, Chess, Swimming, Badminton, Martial-Arts, etc.
- **Multiverses** — RPG, Podcasts, Documentaries, Books, Anime, etc.
- **Natural Languages** — Spanish, Mandarin, Japanese, Punjabi, Tamil!
- **Philosophy** — Ontology, Epistemology, Morality, Ethics, AI-alignment, etc.
- **Traveling** — Stars, SpaceX, NASA, Auroras, Universities, Museums, Monuments, etc.
- **Music** — Rock, Pop, EDM, MeaningWave, LoFi, Rap, etc.
- **Guitar** — Beginner (Eartuning, Staff Notation, Electric, Yousician, etc.)